

Vincent Lin

Seeking Full-Time Engineering Roles | Available May 2027

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EDUCATION

Rochester Institute of Technology, Rochester, NY

Aug 2022 – May 2027

GPA: 3.41 | **Awards:** Uncommon Hacks 2026 Social Track 1st Place, Dean's List, AP Scholar

B.S. - Double Major: Electrical Engineering Technology & Mechatronics Engineering Technology Expected May 2027

EXPERIENCE

Hardware Design Engineer

Jun 2026 – Aug 2026

LoomLogic Inc

New York, NY

- Led the end-to-end redesign of edge-AI textiles classifier by designing and integrating 4 hardware subsystems in SolidWorks
- Validated system design against VOC and functional requirements by conducting beam deflection and lighting/FOV analysis
- Developed complete Excel BOM in 6 weeks under \$3000 by applying DFMA to standardize hardware and simplify assembly
- Finalized production hand-off by commissioning the workstation and documenting an installation manual for assembly

Robotics Automation Field Engineer

Sep 2025 – Dec 2025

Retiina

USPS Atlanta RPDC

- Operated and optimized USPS automated parcel sortation system, supporting 800,000+ daily package throughput
- Led the Operational Excellence team in a DOE-style motor analysis project, reducing system downtime to 8%
- Designed IoT data collection system to gauge motor noise, temperature, and vibration for predictive maintenance
- Troubleshooted and upgraded Allen-Bradley PLCs by reprogramming automation control logic using Studio 5000

Design and Manufacturing Intern

May 2025 – Aug 2025

Electronic Hardware Corporation

Ronkonkoma, NY

- Modeled and prototyped injection-moldable control knobs in Autodesk Inventor for Defense and Aerospace OEMs
- Published 45+ new GD&T-compliant drawings in AutoCAD for bulk manufacturing of control knobs in defense aircraft
- Created 20+ official manufacturing work orders for DFM of custom parts based on in-house manufacturing processes

Makerspace Lab Assistant (Part-Time)

Jan 2024 – Present

RIT SHED

Rochester, NY

- Trained over 2000 students in learning to use FDM/SLA Printers, Laser Cutters, Soldering and Machine Shop Tools

PROJECTS

Polar CNC Sisyphus Table | *VS Code, Python, C++, SolidWorks*

Mar 2026 – Present

- Building SVG to Polar "G-code" slicer using NumPy and coordinate geometry to drive a Polar CNC sand table
- Developed C++ firmware on Nano to drive NEMA17 steppers and used PySerial as data bridge from Python terminal
- Designing 2-axis rotary slider mechanism in SolidWorks and simulating complete motor control system in CircuitDesigner

MPU6050 Gyroscope Controller | *ROS2/micro-ros, C++, PlatformIO, Linux (WSL)*

Jun 2026 – Present

- Simulating gyroscope controller by translating MPU6050 data from ESP32 into quaternions using ROS2 middleware and Rviz

LifeStory | *UncommonHacks '26 1st Place, OpenCV, PyTorch, DINOv2*

devpost.com/software/lifestory

- Developed a photo-based record player that turns physical photos into memory playback for Alzheimer's patients
- Built computer vision pipeline using ESP32-S3 CAM, DINOv2 embeddings, ORB matching, and RANSAC homography
- Tuned hybrid DINOv2 + ORB matching algorithm through multiple edge cases to significantly reduce false positives

MSP432 Object Detection Robot | *Embedded C, TI MSP432, TensorFlow Lite Micro*

Mar 2026 – Jun 2026

- Built real-time object detection system using fine-tuned FOMO model running TensorFlow Lite Micro on ESP32-S3
- Developed bare-metal C firmware for MSP432 motor control, ultrasonic sensing, and inference feedback via UART
- Implemented navigation logic using FSM, interrupt timers, and proportional error between frame center and FOMO output

Automated Blast Gates | *Electrical/Embedded Systems Lead*

Jan 2026 – May 2026

- Optimized vacuum flow and shop safety by designing, manufacturing, and integrating "smart" blast gates with the Wood Shop
- Developed autonomous closed-loop positioning circuit using L298N integrated with Nano logic and linear actuator feedback
- Established ESP32-Nano broadcast architecture via 8N1 UART and devised flow balancing equation for local computation
- Validated reliability over 5,000 actuation cycles against functional spec by using DFMEA results to conduct endurance testing

SKILLS

Software: C, C++, Python, Git, OpenCV, PyTorch, ROS2/micro-ROS, MATLAB, Linux, ABB RAPID, Minitab, Microsoft Office

Electrical: NI Multisim, Altium Designer, ModelSim, Simulink, VHDL, ARM-Cortex M4, ESP32-S3, DE0-CV FPGA, Intel Quartus Prime, Control Theory, Arduino, PlatformIO, ABB RobotStudio, Allen-Bradley PLC, Studio 5000, UART/I2C/SPI

Mechanical: Autodesk Inventor, SolidWorks, Fusion 360, AutoCAD, Laser Cutting, 3D Printing, GD&T, Soldering, Machine Shop Tools, CNC Mill, Lean Six Sigma, Statistical Design of Experiments, DFMEA, MS Project

CERTIFICATIONS

Lean Six Sigma – Yellow Belt | Certified SolidWorks Associate – Mechanical Design | Autodesk Inventor Certified User